Simple Linear Regression

Reading Objectives:

In this reading, you will learn about the estimation procedure by using simple linear regression. Simple linear regression coefficients can be found with the help of the sum of squared errors. Here, you will also learn about the matrix representation of normal equations.

Main Reading Section:

Regression Analysis:

The objective of many statistical investigations is to make predictions which require dependent and independent variables, preferably on the basis of mathematical equations.

Examples:

An engineer may wish to predict the amount of oxide (dependent variable) that will form on the surface of a metal baked in an oven for 1 hour at 200°C.

The amount of deformation of a ring subjected to a compressive force of 1,000 pounds.

The number of miles to wear out a tire as a function of tread thickness and composition.

In regression analysis, a random variable Y (dependent variable or response variable) depends on the value x of the other variable (predictor variable or input variable).

To state the problem formally

We have n paired observations (x_i, y_i) and assume that the regression of y on x is linear.

To find the best fit $\overline{y} = a + b\overline{x}$, where a and b are constants, e_i, i.e., the error in predicting the value of y corresponding to the given x_i, is given by $e_i = y_i - \overline{y}_i$.

To determine a and b so that sum of these errors is as small as possible, i.e., $\sum_{i=0}^{n} e_i = 0$, sometimes the error could be positive or negative.

We shall minimize the sum of the squares of ei (squares of the deviations from the mean in the definition of the standard deviation).

In other words, we apply the **principle of least squares**.

Normal Equations for the Least Square Estimators

A necessary condition that the sum of squared deviations,

$$\sum_{i=1}^{n} (y_i - a - bx_i)^2$$

be a minimum is the vanishing of the partial derivatives with respect to a and b, we thus have,

$$2\sum_{i=1}^{n} [y_i - (a + bx_i)](-1) = 0$$

$$2\sum_{i=1}^{n} [y_i - (a + bx_i)](-x_i) = 0$$

and we can rewrite these two equations as

$$\sum_{i=1}^{n} y_i = an + b\sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i y_i = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2$$